Incremental Data-Driven Robust Crew Pairing Optimization

Under Evolving Uncertainty

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Problem

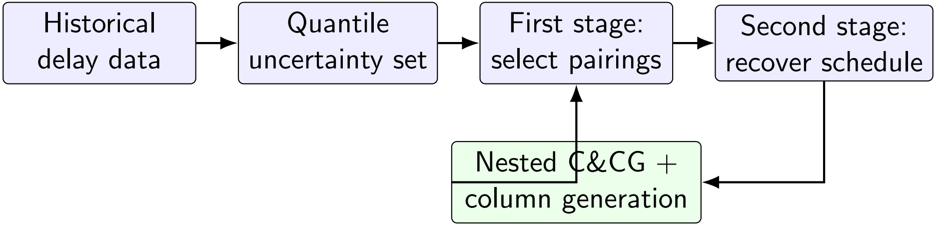
- Airline crew schedules must cover all flights.
- Delays can break planned crew connections.
- Historical data may miss rare but important disruptions.

Main idea

- Build delay bounds from empirical quantiles.
- Model only harmful one-sided disruptions.
- Use a budget parameter to balance robustness and cost.

$$\hat{\mathcal{U}}(\Gamma) = \left\{ \Delta : 0 \leq \Delta_f \leq d_f,\ \sum_f \Gamma_f \leq \Gamma,\ \Delta_f = \Gamma_f d_f \right\}$$

Solution Framework



- First stage: choose feasible crew pairings.
- Second stage: minimize recovery cost under worst-case delays.
- Algorithm adds violated scenarios and negative-reduced-cost pairings.

Conclusion

Main contribution: incremental re-optimization

- New data update the uncertainty bounds.
- If uncertainty shrinks, the previous robust plan remains feasible.
- If uncertainty expands, add new cuts and re-optimize.
- This avoids rebuilding the full model from zero.

Takeaway

The project designs a robust crew pairing method that learns from delay data and can adapt as uncertainty evolves.